

Hein Zarni Aung

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EDUCATION

Stony Brook University — Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Systems Fundamentals I & II, Algorithms, Statistics & Probability

RESEARCH

Undergraduate Research Assistant — Ethos Lab, SBU — Sep 2025 – Dec 2025 (paused)

PI: Abisheka Pitumpe | Team size: 2

- Built a real-time pipeline to monitor newly issued SSL/TLS certificates and flag fraudulent domains, scraping and processing raw HTML, CSS, JS, and URL data from newly created sites.
- Used Selenium to automate certificate/domain scraping and Python to filter and triage flagged sites, designing keyword- and heuristic-based detection logic with a faculty PI.

INTERSHIPS

Software Engineering Intern — Clustr — Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Designed and implemented server-side APIs (Firebase Cloud Functions) handling CRUD operations, permission enforcement, and event-driven workflows (join/leave, activity feeds) for a production social app.
- Contributed 50+ commits across backend and frontend features via pull requests, participating in code review and resolving merge conflicts across feature branches with an 8-person team.

Software Engineering Intern — Prenora Dynamics LLC — Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University — Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation and hosted 2-hour review sessions reaching 100+ students; materials were adopted as the basis for subsequent semesters.

PROJECTS

TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon) — Apr 2026

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built a data pipeline extracting pitch, rhythm, and tone features from audio using Essentia/NumPy to power an AI-matching tool pairing music students with instructors based on technical weaknesses.
- Redesigned a real-time feature-scoring algorithm from a planned ML model to a rule-based approach under strict time and compute constraints, prioritizing correctness and latency within a 36-hour build window.

Discord Voice Channel Activity Monitor — Node.js, Discord.js, Cloud

- Built a real-time, event-driven Node.js service using the Discord.js API to detect and broadcast voice-channel state changes to users with low-latency notifications, supporting customizable DM alerts and per-server/per-user thresholds via JSON-based configuration.
- Deployed on a cloud server using PM2 for process reliability; supported 50 concurrent active users.

SKILLS

Languages: Python, Java, C, TypeScript, JavaScript

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium

Frontend: React, React Native, Next.js